

Лабораторная работа №5

Конфигурация VLAN

Поляков Г. С. true Российский университет дружбы народов, Москва, Россия

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Информация

- Поляков Глеб Сергеевич
- студент
- Российский университет дружбы народов

Настройка Trunk

```
msk-donskaya-gspolyakov-sw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-gspolyakov-sw-1(config)#inter
msk-donskaya-gspolyakov-sw-1(config)#interface g0/1
msk-donskaya-gspolyakov-sw-1(config-if)#swi
msk-donskaya-gspolyakov-sw-1(config-if)#switchport mo
msk-donskaya-gspolyakov-sw-1(config-if)#switchport mode tru
msk-donskaya-gspolyakov-sw-1(config-if)#switchport mode trunk

msk-donskaya-gspolyakov-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

msk-donskaya-gspolyakov-sw-1(config-if)#exit
msk-donskaya-gspolyakov-sw-1(config)#
```

```
msk-donskaya-gspolyakov-sw-1(config)#interface f0/24  
msk-donskaya-gspolyakov-sw-1(config-if)#switchport mode trunk  
msk-donskaya-gspolyakov-sw-1(config-if)#exit
```

```
msk-donskaya-gspolyakov-sw-1(config)#interface g0/2
msk-donskaya-gspolyakov-sw-1(config-if)#switchport mode trunk

msk-donskaya-gspolyakov-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet

msk-donskaya-gspolyakov-sw-1(config-if)#exit
```

```
msk-donskaya-gspolyakov-sw-1(config)#interface f0/1
msk-donskaya-gspolyakov-sw-1(config-if)#
msk-donskaya-gspolyakov-sw-1(config-if)#switchport mode trunk
msk-donskaya-gspolyakov-sw-1(config-if)#exit
msk-donskaya-gspolyakov-sw-1(config)#
```



```
msk-donskaya-gspolyakov-sw-2(config)#interface g0/1
msk-donskaya-gspolyakov-sw-2(config-if)#sw
msk-donskaya-gspolyakov-sw-2(config-if)#switchport m
msk-donskaya-gspolyakov-sw-2(config-if)#switchport mode t
msk-donskaya-gspolyakov-sw-2(config-if)#switchport mode trunk
msk-donskaya-gspolyakov-sw-2(config-if)#exit
```

```
msk-donskaya-gspolyakov-sw-2(config)#interface g0/2
msk-donskaya-gspolyakov-sw-2(config-if)#switchport mode trunk

msk-donskaya-gspolyakov-sw-2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0
msk-donskaya-gspolyakov-sw-2(config-if)#exit
```

```
msk-donskaya-gspolyakov-sw-3(config)#interface g0/1
msk-donskaya-gspolyakov-sw-3(config-if)#swi
msk-donskaya-gspolyakov-sw-3(config-if)#switchport m
msk-donskaya-gspolyakov-sw-3(config-if)#switchport mode tr
msk-donskaya-gspolyakov-sw-3(config-if)#switchport mode trunk
msk-donskaya-gspolyakov-sw-3(config-if)#exit
```

```
msk-donskaya-gspolyakov-sw-4(config)#interface g0/1
msk-donskaya-gspolyakov-sw-4(config-if)#sw
msk-donskaya-gspolyakov-sw-4(config-if)#switchport mo
msk-donskaya-gspolyakov-sw-4(config-if)#switchport mode tr
msk-donskaya-gspolyakov-sw-4(config-if)#switchport mode trunk
msk-donskaya-gspolyakov-sw-4(config-if)#exit
```

```
msk-pavlovskaya-gspolyakov-sw-1(config)#interface f0/24
msk-pavlovskaya-gspolyakov-sw-1(config-if)#sw
msk-pavlovskaya-gspolyakov-sw-1(config-if)#switchport m
msk-pavlovskaya-gspolyakov-sw-1(config-if)#switchport mode t
msk-pavlovskaya-gspolyakov-sw-1(config-if)#switchport mode trunk
msk-pavlovskaya-gspolyakov-sw-1(config-if)#exit
```

Настройка Сервера VLAN

```
msk-donskaya-gspolyakov-sw-1(config)#vtp mode server
Device mode already VTP SERVER.
msk-donskaya-gspolyakov-sw-1(config)#vtp pass
msk-donskaya-gspolyakov-sw-1(config)#vtp password cisco
%The VTP password cannot be set for NULL domain
msk-donskaya-gspolyakov-sw-1(config)#vtp domain donskaya
Changing VTP domain name from NULL to donskaya
msk-donskaya-gspolyakov-sw-1(config)#vtp password cisco
Setting device VLAN database password to cisco
msk-donskaya-gspolyakov-sw-1(config)#vlan 2
msk-donskaya-gspolyakov-sw-1(config-vlan)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-donskaya-gspolyakov-sw-1(config-vlan)#nam
msk-donskaya-gspolyakov-sw-1(config-vlan)#name management
msk-donskaya-gspolyakov-sw-1(config-vlan)#vlan 3
msk-donskaya-gspolyakov-sw-1(config-vlan)#name servers
msk-donskaya-gspolyakov-sw-1(config-vlan)#vlan 101
msk-donskaya-gspolyakov-sw-1(config-vlan)#name dk
msk-donskaya-gspolyakov-sw-1(config-vlan)#vlan 102
msk-donskaya-gspolyakov-sw-1(config-vlan)#name deparaments
msk-donskaya-gspolyakov-sw-1(config-vlan)#vlan 103
msk-donskaya-gspolyakov-sw-1(config-vlan)#name adm
msk-donskaya-gspolyakov-sw-1(config-vlan)#vlan 104
msk-donskaya-gspolyakov-sw-1(config-vlan)#name other
msk-donskaya-gspolyakov-sw-1(config-vlan)#exit
```

Настройка Клиента VLAN

```
msk-donskaya-gspolyakov-sw-2(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-gspolyakov-sw-2(config)#interface range f0/1 - 2
msk-donskaya-gspolyakov-sw-2(config-if-range)#swi
msk-donskaya-gspolyakov-sw-2(config-if-range)#switchport mo
msk-donskaya-gspolyakov-sw-2(config-if-range)#switchport mode a
msk-donskaya-gspolyakov-sw-2(config-if-range)#switchport mode access
msk-donskaya-gspolyakov-sw-2(config-if-range)#sw
msk-donskaya-gspolyakov-sw-2(config-if-range)#switchport ac
msk-donskaya-gspolyakov-sw-2(config-if-range)#switchport access v
msk-donskaya-gspolyakov-sw-2(config-if-range)#exit
msk-donskaya-gspolyakov-sw-2(config)#interface range f0/1 - 24
msk-donskaya-gspolyakov-sw-2(config-if-range)#switchport mode access
msk-donskaya-gspolyakov-sw-2(config-if-range)#switchport ac
msk-donskaya-gspolyakov-sw-2(config-if-range)#switchport access vl
msk-donskaya-gspolyakov-sw-2(config-if-range)#switchport access vlan 3
msk-donskaya-gspolyakov-sw-2(config-if-range)#exit
msk-donskaya-gspolyakov-sw-2(config)#exit
```

```
msk-donskaya-gspolyakov-sw-3(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-gspolyakov-sw-3(config)#int
msk-donskaya-gspolyakov-sw-3(config)#interface r
msk-donskaya-gspolyakov-sw-3(config)#interface range f0/1 - 2
msk-donskaya-gspolyakov-sw-3(config-if-range)#sw
msk-donskaya-gspolyakov-sw-3(config-if-range)#switchport m
msk-donskaya-gspolyakov-sw-3(config-if-range)#switchport mode a
msk-donskaya-gspolyakov-sw-3(config-if-range)#switchport mode access
msk-donskaya-gspolyakov-sw-3(config-if-range)#s
msk-donskaya-gspolyakov-sw-3(config-if-range)#sw
msk-donskaya-gspolyakov-sw-3(config-if-range)#switchport a
msk-donskaya-gspolyakov-sw-3(config-if-range)#switchport access v
msk-donskaya-gspolyakov-sw-3(config-if-range)#switchport access vlan 3
msk-donskaya-gspolyakov-sw-3(config-if-range)#exit
```

```
msk-donskaya-gspolyakov-sw-4(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-gspolyakov-sw-4(config)#inter
msk-donskaya-gspolyakov-sw-4(config)#interface ra
msk-donskaya-gspolyakov-sw-4(config)#interface range f0/1 - 5
msk-donskaya-gspolyakov-sw-4(config-if-range)#sw
msk-donskaya-gspolyakov-sw-4(config-if-range)#switchport ac
msk-donskaya-gspolyakov-sw-4(config-if-range)#switchport mode access
msk-donskaya-gspolyakov-sw-4(config-if-range)#switchport acc
msk-donskaya-gspolyakov-sw-4(config-if-range)#switchport access v
msk-donskaya-gspolyakov-sw-4(config-if-range)#switchport access vlan 101
msk-donskaya-gspolyakov-sw-4(config-if-range)#interface range f0/6 - 10
msk-donskaya-gspolyakov-sw-4(config-if-range)#switchport mode access
msk-donskaya-gspolyakov-sw-4(config-if-range)#switchport access vlan 102
msk-donskaya-gspolyakov-sw-4(config-if-range)#interface range f0/11 - 15
msk-donskaya-gspolyakov-sw-4(config-if-range)#switchport mode access
msk-donskaya-gspolyakov-sw-4(config-if-range)#switchport access vlan 103
msk-donskaya-gspolyakov-sw-4(config-if-range)#interface range f0/16 - 24
msk-donskaya-gspolyakov-sw-4(config-if-range)#switchport mode access
msk-donskaya-gspolyakov-sw-4(config-if-range)#switchport access vlan 104
msk-donskaya-gspolyakov-sw-4(config-if-range)#exit
msk-donskaya-gspolyakov-sw-4(config)#exit
```

```
msk-pavlovskaya-gspolyakov-sw-1(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-pavlovskaya-gspolyakov-sw-1(config)#in
msk-pavlovskaya-gspolyakov-sw-1(config)#interface ra
msk-pavlovskaya-gspolyakov-sw-1(config)#interface range f0/1 - 15
msk-pavlovskaya-gspolyakov-sw-(config-if-range)# sw
msk-pavlovskaya-gspolyakov-sw-(config-if-range)# switchport m
msk-pavlovskaya-gspolyakov-sw-(config-if-range)# switchport mode a
msk-pavlovskaya-gspolyakov-sw-(config-if-range)# switchport mode access
msk-pavlovskaya-gspolyakov-sw-(config-if-range)# switchport ac
msk-pavlovskaya-gspolyakov-sw-(config-if-range)# switchport access v
msk-pavlovskaya-gspolyakov-sw-(config-if-range)# switchport access vlan 101
msk-pavlovskaya-gspolyakov-sw-(config-if-range)#interface f0/24
msk-pavlovskaya-gspolyakov-sw-1(config-if)#exit
msk-pavlovskaya-gspolyakov-sw-1(config)#interface f0/24
msk-pavlovskaya-gspolyakov-sw-1(config-if)#sw
msk-pavlovskaya-gspolyakov-sw-1(config-if)#switchport mo
msk-pavlovskaya-gspolyakov-sw-1(config-if)#switchport mode a
msk-pavlovskaya-gspolyakov-sw-1(config-if)#switchport mode access
msk-pavlovskaya-gspolyakov-sw-1(config-if)%%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/24 VLAN1.

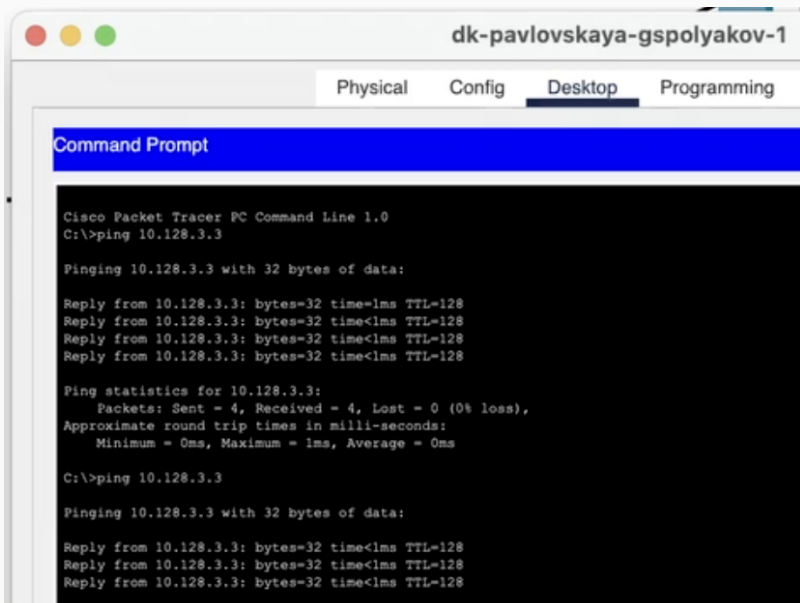
%%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/24 on VLAN0001. Inconsistent port type.

msk-pavlovskaya-gspolyakov-sw-1(config-if)#|
```

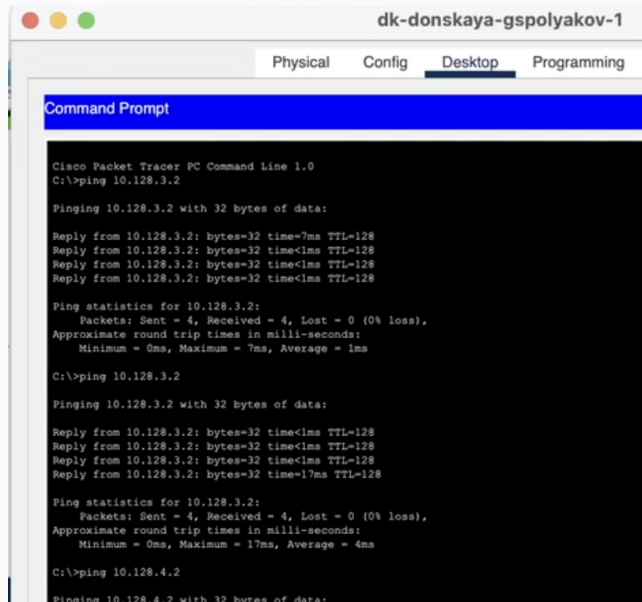
```
msk-pavlovskaya-gspolyakov-sw-1(config-if)#interface f0/20
msk-pavlovskaya-gspolyakov-sw-1(config-if)#sw
msk-pavlovskaya-gspolyakov-sw-1(config-if)#switchport m
msk-pavlovskaya-gspolyakov-sw-1(config-if)#switchport mode a
msk-pavlovskaya-gspolyakov-sw-1(config-if)#switchport mode access
msk-pavlovskaya-gspolyakov-sw-1(config-if)#sw
msk-pavlovskaya-gspolyakov-sw-1(config-if)#switchport a
msk-pavlovskaya-gspolyakov-sw-1(config-if)#switchport access v
msk-pavlovskaya-gspolyakov-sw-1(config-if)#switchport access vlan 104
msk-pavlovskaya-gspolyakov-sw-1(config-if)#exit
msk-pavlovskaya-gspolyakov-sw-1(config)#exit
```

Ping

Ping



Ping



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'dk-donskaya-gspolyakov-1'. The 'Desktop' tab is selected. The Command Prompt displays the output of two ping commands. The first command is 'ping 10.128.3.2', which shows four successful replies with 32 bytes each, times less than 1ms, and a TTL of 128. The statistics show 4 packets sent, 4 received, 0 lost, and an average round trip time of 1ms. The second command is 'ping 10.128.3.2' again, showing four successful replies with 32 bytes each, times less than 1ms, and a TTL of 128. The statistics show 4 packets sent, 4 received, 0 lost, and an average round trip time of 4ms. The third command is 'ping 10.128.4.2', which is partially visible at the bottom of the screen.

```
dk-donskaya-gspolyakov-1
Physical Config Desktop Programming
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.3.2

Pinging 10.128.3.2 with 32 bytes of data:

Reply from 10.128.3.2: bytes=32 time<1ms TTL=128
Reply from 10.128.3.2: bytes=32 time<1ms TTL=128
Reply from 10.128.3.2: bytes=32 time<1ms TTL=128
Reply from 10.128.3.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.128.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 1ms

C:\>ping 10.128.3.2

Pinging 10.128.3.2 with 32 bytes of data:

Reply from 10.128.3.2: bytes=32 time<1ms TTL=128
Reply from 10.128.3.2: bytes=32 time<1ms TTL=128
Reply from 10.128.3.2: bytes=32 time<1ms TTL=128
Reply from 10.128.3.2: bytes=32 time=17ms TTL=128

Ping statistics for 10.128.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 17ms, Average = 4ms

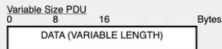
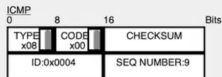
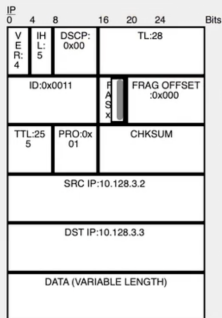
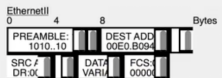
C:\>ping 10.128.4.2

Pinging 10.128.4.2 with 32 bytes of data:
```


ICMP

ICMP

PDU Formats



ICMP

